

Machine Learning Project
Group Number-8
Submitted by
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overview:

This project focuses on building a binary classification model based on the Florida 2013 mortgage dataset. The goal is to predict whether a loan applicant will be approved or not, based on various personal, financial, and application-specific features. The dataset includes both numerical and categorical variables, some of which are anonymized. The project is divided into several stages, beginning with data exploration and visualization, followed by feature selection, model training, evaluation, and interpretation. Throughout the project, we apply common machine learning methodologies and data science practices to ensure that the model is well-understood, robust, and interpretable. The insights gained from the early stages of the project are essential for building a predictive model that captures the key patterns in the data and generalizes well to new applicants.

Part 1: Data Exploration

In this section, we conducted a comprehensive exploration of the training dataset to understand the characteristics of each feature, identify patterns, detect anomalies, and prepare the data for future modeling. We began by distinguishing between categorical and numerical features, then analyzed their distribution, frequency, and relationship with the target variable (label). The goal was to extract meaningful insights from the data, supported by both statistics and visualization.

We identified that the dataset contains several numerical features, such as `years_of_experience`, `prev_salary`, and anonymized columns A, B, C, and D. Additionally, it contains categorical variables like `country`, `sex`, `worked_in_the_past`, `disability`, and `mental_issues`, among others. Each categorical feature was visualized using bar charts or pie charts, depending on the number of unique values, while numerical features were presented using histograms and boxplots to assess their distribution and detect potential outliers.

Descriptive statistics such as mean, median, standard deviation, and range were computed for numerical features. We found that several variables contain missing values and a few include extreme values that may require special attention during preprocessing. For instance, `prev_salary` and `years_of_experience` have highly skewed distributions, indicating potential outliers or long tails.

To better understand relationships among features, we computed a Pearson correlation matrix and visualized it using a heatmap. This analysis revealed several significant relationships, which are summarized below:

- C & population (0.88) and C & number_of_owner_occupied_units (0.95):
Feature "C" shows strong positive correlations with both population and the number of owner-occupied units, suggesting that it may reflect regional or housing density patterns.
- applicant_income & loan_amount (0.45):
Applicants with higher income tend to request or qualify for larger loans, which aligns with common financial expectations.

- label & number_of_owner_occupied_units (0.57):
Areas with more owner-occupied housing units are associated with a higher likelihood of label = 1, indicating a potential socio-economic influence on approval outcomes.
- A & tract_to_msamd_income (0.33):
A moderate correlation suggests that feature "A" may be related to regional income levels or geographical income disparity.
- label & tract_to_msamd_income (0.084):
A weak correlation that may hint at a slight income advantage, though it is likely not practically significant.

These correlation insights provide important context for selecting and transforming features. They also highlight redundancy in certain variables, the presence of hidden relationships in anonymized features, and signals that may improve model performance. Overall, the exploration phase helped us shape a clear understanding of the dataset and form hypotheses that will guide us in feature selection and model training.

PART 2 : Data Preparation and Preprocessing

First we chose to delete the ID column because it acts only as a unique identifier and holds no predictive value. Keeping it could mislead machine learning models into treating it as a meaningful numeric feature, despite having no correlation with the target.

האם קיימים נתונים חריגים outliers בדאטה ? אם כן , עליכם להסירם או לפחות לתת עליהם את הדעת :

During the analysis, we identified outliers in several numerical features—particularly income, population size, and loan amount—which were clearly visible in the boxplots. Since outliers can distort model performance and introduce bias, we applied Interquartile Range (IQR) capping to replace extreme values with boundary values (25th or 75th percentile) rather than removing entire rows. This approach preserved the overall data structure, reduced the influence of outliers, and maintained the richness and variability of the dataset. After capping, the updated boxplots showed cleaner, more balanced distributions without altering the general shape of the data.

האם הנתונים מנורמלים ? אם לא , האם צריך לנרמל אותם ? מה החשיבות של נרמול הנתונים בבעיה ?

Yes, the data was normalized to address skewed distributions and inconsistent feature scales. We first applied a logarithmic transformation to long-tailed features such as income and population to reduce right skewness. Then, we used Min-Max scaling to bring all numerical features into a common range between 0 and 1. This two-step process ensured that features with large numeric ranges, like income (in the thousands) and minority population (as a percentage), did not dominate the learning process or bias the models.

Normalization was essential for several reasons. It improved learning for models using gradient-based optimization (like logistic regression and neural networks), ensured fairer distance

calculations for models like K-Nearest Neighbors, and prepared the data for dimensionality reduction techniques such as PCA, which are sensitive to variance. By normalizing the data, we improved both model performance and stability while maintaining fairness across features, making the dataset more suitable for advanced modeling.

האם ישנם נתונים חסירים ? כיצד בחרתם לטפל בהם ומדוע באופן זה ?

The dataset initially contained missing values across several columns. Four features—applicant_race_name_2, applicant_race_name_3, co_applicant_race_name_2, and co_applicant_race_name_3—had over 99% missing values and were therefore dropped, as they contributed little value and added unnecessary noise. For the remaining features, we used imputation: missing numerical values were filled with the median to reduce the impact of outliers, while categorical values were filled with the most frequent category (mode) to maintain consistency. This simple yet effective approach preserved the dataset's structure, avoided the loss of rows, and ensured that imputed values reflected the original distribution—leaving us with a complete and model-ready dataset.

התמודדות עם משתנים קטגוריאליים :

To prepare the dataset for machine learning, we applied One-Hot Encoding to the categorical variables. This method creates binary columns for each category, allowing models to interpret categorical data without assuming any ordinal relationship. One-Hot Encoding is broadly compatible with most machine learning algorithms and enhances both model performance and interpretability. However, the county feature posed a challenge due to its high cardinality (67 unique values), which would have led to a large, sparse matrix and increased risk of overfitting.

To address this, we grouped counties into three broader regions—North Florida, Central Florida, and South Florida—based on geographic location. This reduced dimensionality while preserving meaningful spatial information. We also dropped the msamd column, which was redundant since each county corresponded to a unique MSA. After these adjustments, One-Hot Encoding was applied to the remaining categorical features such as loan_type, applicant_ethnicity, and lien_status, ensuring the data was fully numeric and suitable for modeling without introducing artificial relationships.

האם המימדיות של הבעיה גדולה מדי ? למה מימדיות גדולה עלולה ליצור בעיה ? איך נזהה כי מימדיות הבעיה גדולה מדי ?

Yes, we encountered a high dimensionality issue during preprocessing. After applying One-Hot Encoding, the number of features in our dataset increased sharply from 32 to 81 columns. This substantial rise in dimensionality can negatively affect model performance by introducing redundancy, increasing computational load, and making the dataset more complex to analyze and interpret.

High dimensionality becomes problematic when the number of features is large relative to the number of observations, as it can lead to overfitting, reduced generalization, and the "curse of dimensionality," where data becomes sparse and patterns become harder to detect. It can also result in irrelevant or duplicate information being included, making the model more prone to noise and less

interpretable. In our case, the nearly threefold increase in feature count indicated a need to apply dimensionality reduction techniques—especially since some models, like linear classifiers, are more sensitive to high-dimensional input than others, such as tree-based methods.

part 3 : modeling

In this part , we implemented and evaluated four classification models to predict the binary target variable. We selected two basic models: Logistic Regression and K-Nearest Neighbors (KNN), and two advanced models: Random Forest and Multi-Layer Perceptron (MLP)

For the Initial models:

- Logistic Regression was chosen for its simplicity and interpretability. It uses a linear decision boundary and is useful as a baseline. We relied on the default hyperparameters, and regularization was handled automatically.
- K-Nearest Neighbors (KNN) was used with $k=5$. This hyperparameter (`n_neighbors`) determines how many neighbors are considered when predicting the class of a point. A smaller k leads to lower bias but higher variance.

For the advanced models:

- Random Forest was used for both modeling and feature selection. We used 100 estimators (`n_estimators=100`) and the default maximum depth. The model combines the predictions of multiple decision trees, reducing overfitting and increasing generalization. Important features were selected based on feature importances calculated by the model.
- MLP (Multi-Layer Perceptron) was implemented using `hidden_layer_sizes=(100,)`, a single hidden layer with 100 neurons. We used the ReLU activation and Adam optimizer. Key hyperparameters that impact performance include the learning rate, number of epochs, and network architecture, which balance bias and variance.

In terms of feature importance, the Random Forest model provided insight into which features were most informative for classification. Features with the highest importance scores were retained during dimensionality reduction. This step contributed to model interpretability and computational efficiency.

Overall, this modeling phase helped us compare different algorithms, optimize their parameters, and reduce dimensionality while preserving predictive power.

part 4 : Model Evaluation

עליכם להסביר התאים בתוך המטריצה בהקשר למודל שבחרתם , כלומר מה ניתן להסיק על ביצוע המודל בהקשר זה:

We evaluated the Random Forest model using a Confusion Matrix, which showed how well it distinguishes between “Approved” and “Not Approved” applications. The model correctly identified **23,155** “Not Approved” cases (TN) and **23,106** “Approved” cases (TP), while misclassifying **6,212** false approvals (FP) and **6,260** false rejections (FN). These values highlight both the model's strengths and potential areas for error.

The model achieved an overall accuracy of **78.8%**, indicating balanced performance across both classes. Its low number of false positives is especially important in avoiding unqualified loan approvals. This suggests the model is reliable and suitable for real-world application.

Across all models, performance on the validation set was similar to or better than on the training set, suggesting no significant overfitting. For example, the MLP Classifier performed better on validation (81.44%) than training (77.22%), and Random Forest showed a similar trend. To further improve generalization, we used validation splitting, tested multiple models, and could apply techniques like cross-validation, hyperparameter tuning, and feature selection.

part 6 :

In Part 3 of the project, we implemented and evaluated several machine learning models,

including XGBoost, which we used as one of our advanced models. Although XGBoost was not covered in the course material, we chose to use it due to its strong performance and popularity in practical applications.

We trained the XGBoost classifier using the preprocessed dataset, tuning key hyperparameters such as the number of estimators and learning rate. The model was evaluated using metrics like accuracy, AUC, and the ROC curve. XGBoost produced high predictive performance and provided valuable insights through feature importance analysis.

Its ability to handle missing values, reduce overfitting via regularization, and effectively model complex patterns made it a suitable choice for our classification task.

summary

In this project, we developed a binary classification model using the Florida 2013 mortgage dataset to predict loan approval outcomes. We began with thorough data exploration and visualization to understand feature distributions and correlations. During preprocessing, we addressed missing values, handled outliers, normalized features, and reduced high dimensionality through encoding and grouping. We implemented four main models—Logistic Regression, K-Nearest Neighbors, Random Forest, and Multi-Layer Perceptron—as well as XGBoost as an additional advanced model. Each model was evaluated using metrics such as accuracy, AUC, and ROC curves, with cross-validation applied to ensure generalization. Finally, we built a complete prediction pipeline and generated probability outputs for the test set. This project showcased the end-to-end machine learning process, from raw data to final predictions, with a strong emphasis on model performance, interpretability, and data-driven decision-making.

נספח העבודה :

Mira Hayat Contribution :

Preprocessing: Converted columns to numeric types, removed outliers, applied normalization, and handled missing values.

Model Evaluation: Built and analyzed confusion matrices, ROC curves, and applied K-Fold Cross Validation to evaluate model performance.

Prediction: Built the final pipeline and made predictions on the test set, saving the results in the required CSV format.

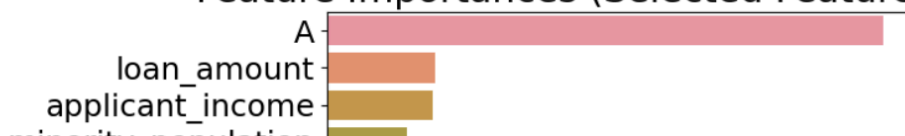
Kamelia Atwan Contribution :

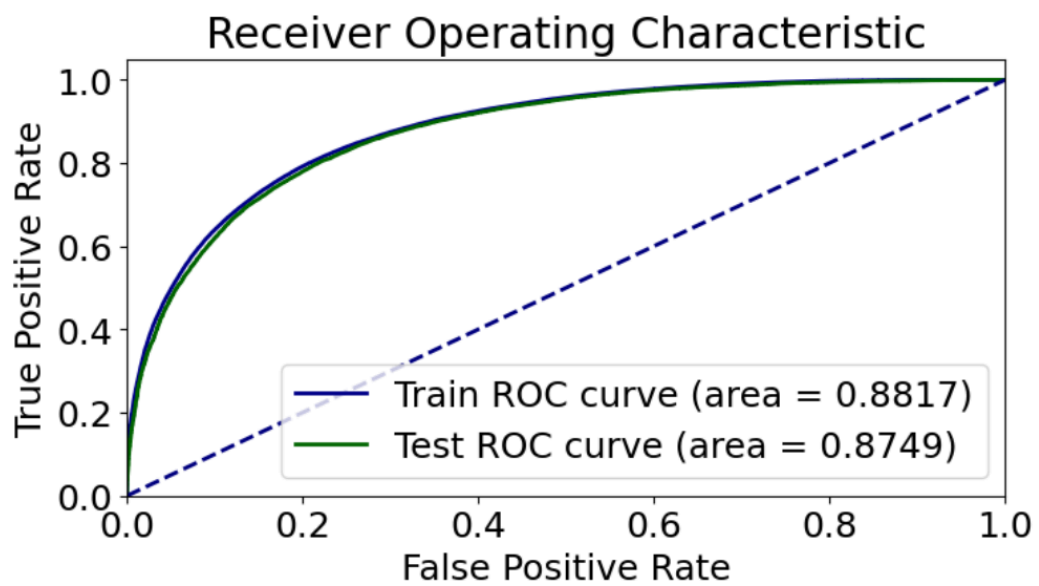
Data Exploration: Examined data types, missing values, and descriptive statistics. Created visualizations including bar plots, pie charts, and boxplots.

Modeling: Implemented four classification models — two basic models (Logistic Regression and K-Nearest Neighbors) and two advanced models (Decision Tree and Artificial Neural Network).

Unfamiliar Method (Part 6): Applied XGBoost as an additional advanced model not covered in the course, explained its contribution, and evaluated its impact on model performance.

Feature Importances (Selected Features Only)





Correlation Matrix of Numerical Features and the label

	loan_amount	county_code	census_tract_number	applicant_income	population	minority_population	hud_median_family_income	tract_to_msamd_income	number_of_owner_occupied_units	A
loan_amount	1	0.001	-0.0073	0.45	-0.0027	-0.014	0.022	0.072	-0.00097	0.019
county_code	0.001	1	0.098	0.0009	0.05	-0.058	-0.2	0.033	0.062	0.01
census_tract_number	-0.0073	0.098	1	-0.0011	0.043	-0.14	-0.27	-0.05	0.14	-0.0083
applicant_income	0.45	0.0009	-0.0011	1	-0.011	-0.028	0.02	0.079	-0.0037	0.014
population	-0.0027	0.05	0.043	-0.011	1	0.14	0.031	0.059	0.87	0.018
minority_population	-0.014	-0.058	-0.14	-0.028	0.14	1	-0.18	-0.27	-0.14	-0.05
hud_median_family_income	-0.022	-0.2	-0.27	0.02	0.031	-0.18	1	-0.0051	0.056	0.033
tract_to_msamd_income	-0.072	0.033	-0.05	0.079	0.059	-0.27	-0.0051	1	0.15	0.044
number_of_owner_occupied_units	-0.00097	0.062	0.14	-0.0037	0.87	-0.14	0.056	0.15	1	0.033
A	-0.019	0.01	-0.0083	0.014	0.018	-0.05	0.033	0.044	0.033	1

Random Forest Confusion Matrix (Counts + Percentages)

